# The POWHEG spectra are scaled event counts

Recovering the counts and the scale factor of every spectrum

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**Summary**

1. Every spectrum, training and test, is *exactly* $t_k = q_s n_k$ with one scale factor $q_s$ per spectrum and integer counts $n_k$; the largest deviation of any $t_k/q_s$ from an integer is $4 \times 10^{-4}$, the precision of the single-precision numbers in the files.
2. The noise variance is therefore $q_s t_k$ exactly; the global $N_{\text{eff}} = 38{,}024$ is no longer needed. It overstated the variance by 2.7% for a typical spectrum, and understated it by a factor 2.5 for the two short runs in the training file (IDs 1814 and 4284, about 15,000–16,000 events).
3. Every run generated 40,000 events (confirmed by the colleague who produced them); the events missing from the binned range fall below 255 or above 1455 GeV. The two short spectra (IDs 1814 and 4284) do not fit: they would need about 24,000 events outside the range, while their first and last bins suggest a few hundred; their normalisation is nevertheless right, so they are most likely runs of about 16,000 events.
4. There is no sign of common random numbers between spectra.
5. New module `dihiggs/counts.py`; it processes the training file in 2 seconds.


## 1 Method

If all events of a run carry the weight $q_s$, the cross sections of the bins of spectrum $s$ are integer multiples of $q_s$, and $q_s$ is their greatest common divisor. Let $t_{\min}$ be the smallest non-zero value. If that bin holds $c$ events, every ratio $c\,t_k/t_{\min}$ is an integer. We try $c = 1, 2, \ldots$ and keep the first $c$ for which all ratios are within 0.01 of an integer; then $n_k = \text{round}(c\,t_k/t_{\min})$ and $q_s = \sum_k t_k / \sum_k n_k$. The first $c$ gives the largest $q_s$; a smaller one ($q_s/2$, say) would be selected instead only if every count of the spectrum were even, which for about 80 bins does not happen. The count $c$ of the smallest bin ranges from 1 (in 15% of the spectra) to 177, with a median of 14. A spurious, smaller $c$ would require about 80 ratios to fall within 0.01 of an integer by chance, a probability of about $0.02^{80}$.

## 2 Results

- **All spectra are exact multiples.** Training: 3,989 of 3,989; test: 1,000 of 1,000. Largest deviation $4.0 \times 10^{-4}$ of a count, for counts up to about 8,000, i.e. a relative precision of $5 \times 10^{-8}$: single-precision rounding.

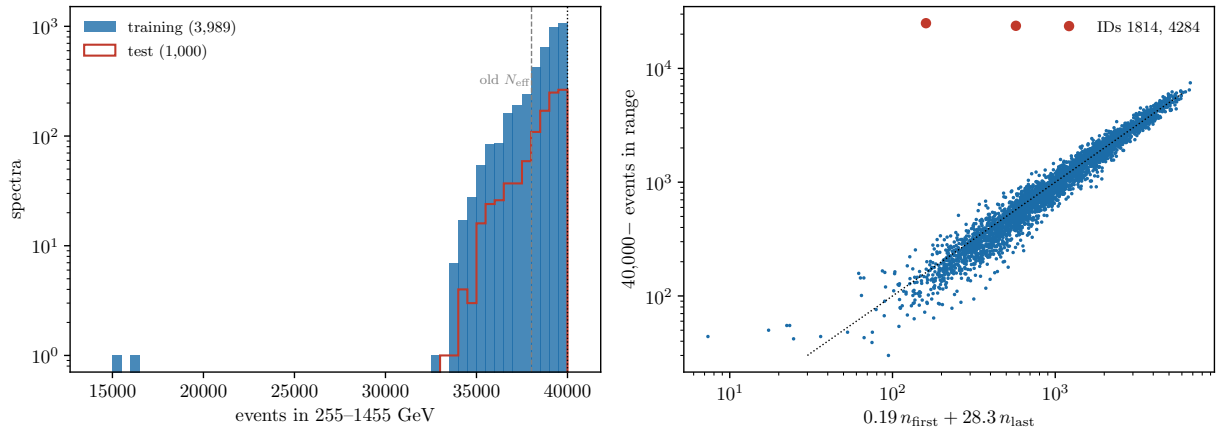



**Figure 1:** Left: the number of events inside the binned range, for the training (filled) and test (line) spectra; the dotted line marks 40,000, the dashed line the old $N_{\text{eff}}$. Right: the number of events missing from 40,000 against a simple proxy for the number expected outside the range (a fit of $a\,n_{\text{first}} + b\,n_{\text{last}}$, the counts in the first and last bins); correlation 0.986. The two short runs (red) are far off the line.

- **Events in range** (fig. 1, left): median 39,043 (training) and 39,041 (test); 5% of the training spectra have fewer than 36,061. The largest is 39,970. Two training spectra, IDs 1814 and 4284, have only 15,031 and 16,310 events. If every run generated 40,000 events, about 24,000 of theirs are unaccounted for. Their normalisation is right, however: their total cross sections agree with the ensemble's prediction at their HEFT points to 0.5% and 0.2%. So these runs most likely produced only about 16,000 events and were normalised to the events they had. They are valid histograms with their own $q_s$, which the method finds automatically; only their noise is larger.
- **40,000 generated events.** If every complete run generated 40,000 events, the deficit $40{,}000 - N_{\text{in}}$ is the number of events below 255 or above 1455 GeV. A crude proxy built from the first and last bins tracks it with correlation 0.986 (fig. 1, right), with no spectrum above 40,000, as expected: every run generated 40,000 events (confirmed by the colleague who produced the samples).
- **Empty bins.** 1,814 bins with no events were dropped from the training file (the test file keeps them as zeros). `add_empty_bins` restores them.

## 3 Independence

**Within a spectrum.** Counts with a fixed total are multinomial, not independent (every run generated 40,000 events): $\text{Cov}(n_i, n_j) = -N p_i p_j$, a correlation of about $-\sqrt{p_i p_j}$, at most a few per cent here; the counts in range are multinomial together with the out-of-range events. A multinomial likelihood equals a product of independent Poisson likelihoods with the normalisation of the spectrum left free; treating the bins as independent Poisson therefore overstates only the information on each spectrum's total, by one count's worth among about 80 bins. We neglect it. The scale factor $q_s = \sigma_{\text{tot}}/N_{\text{gen}}$ also carries POWHEG's integration error on $\sigma_{\text{tot}}$, common to all bins of a spectrum; its size is not known to us.

**Between spectra.** Independence requires a different random seed for every run. Using the difference of each bin from the mean of its two neighbours, a model-free estimate of its fluctuation, the correlation between pairs of test spectra over the 58 bins above 555 GeV averages 0.0006; there is no common fluctuation.

## 4 Consequences

| fine-tuned bootstrap ensemble | s.d. of pull | | | $\langle\chi^2\rangle$ per bin | |
|---|---|---|---|---|---|
| | all | 255–555 GeV | $>$ 555 GeV | 255–555 GeV | $>$ 555 GeV |
| old variance, $t_k\sigma_{\text{tot}}/N_{\text{eff}}$ | 1.069 | 1.145 | 1.042 | 1.62 | 1.10 |
| exact variance, $q_s t_k$ | 1.077 | 1.153 | 1.050 | 1.65 | 1.12 |

**Table 1:** The test-spectrum results of the appendix with the exact variances. The ratio of old to exact variance on the test spectra has median 1.027 and ranges from 0.88 to 1.05.

- The appendix numbers move by about 1%; the model-error coefficient will rise slightly. The noise model was, if anything, too generous, so the model error is not an artefact of an underestimated noise.
- For the Laplace approximation and any Poisson likelihood, the weight of a bin is $1/(q_s f_k)$, with $f_k$ the prediction (`poisson_variance(df, prediction=f)`), which also covers empty bins and bins with a single event.
- The training noise variance $(t_k + u_k)u_k$ of the fine-tune becomes $q_s\,(t_k + q_s)$, or $q_s f_k$; the fine-tune itself need not be redone for the paper, since its result is validated on the test spectra.

## 5 Code

```
from dihiggs import counts
out, summary = counts.recover_counts(df)        # adds 'count' and 'quantum'
full = counts.add_empty_bins(out)               # 80 bins per spectrum
var = counts.poisson_variance(full)             # q * t
var_f = counts.poisson_variance(full, prediction=f)   # q * f
n, q, dev = counts.recover_counts_wide(F)       # test file: one row per spectrum
```